FIITJEE **JEE (Advanced)-2018**

ALL INDIA TEST SERIES

PART TEST – II

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 186**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into Two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. ***Section-A (01– 05, 19 – 23, 37 - 41)*** *contains 15 multiple choice questions which have* ***one correct*** *answer. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

   Section-A (06 – 13, 24 – 31, 42 - 49) *contains 24 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer.*

   Partial Marks **+1** for each correct option provided no incorrect options is selected.

2. ***Section-C (14 – 18, 32 – 36, 50 – 54)*** *contains 15 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A
(One Options Correct Type)

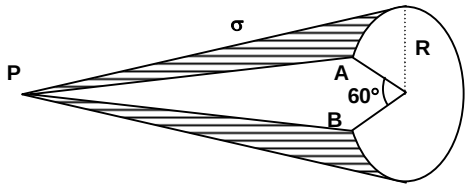
This section contains **5 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. Magnetic field exist in the space and given as $\vec{B} = -\frac{B_0}{\ell^2}x^2\hat{k}$, where B_0 and ℓ are positive constants. A particle having positive charge 'q' and mass 'm' is projected with speed 'v_0' along positive x-axis from the origin. What is the maximum distance of the charged particle from the y-axis before it turns back due to the magnetic field. (Ignore any interaction other than magnetic field)

(A) $\left(\frac{m\ell^2 v_0}{3qB_0}\right)^{1/3}$ (B) $\left(\frac{3m\ell^2 v_0}{2qB_0}\right)^{1/3}$

(C) $\left(\frac{3m\ell^2 v_0}{qB_0}\right)^{1/3}$ (D) $\left(\frac{m\ell^2 v_0}{qB_0}\right)^{1/3}$

2. A non-conducting hollow cone has charge density σ. A part ABP is cut and removed from the cone. The potential due to the remaining portion of the cone at point 'P' is

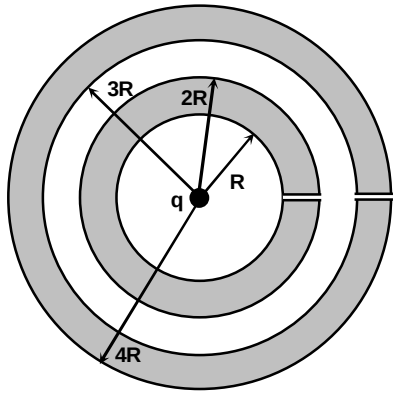


(A) $\frac{5}{6}\frac{\sigma R}{\varepsilon_0}$ (B) $\frac{5}{24}\frac{\sigma R}{\varepsilon_0}$

(C) $\frac{5}{3}\frac{\sigma R}{\varepsilon_0}$ (D) $\frac{5}{12}\frac{\sigma R}{\varepsilon_0}$

Space for Rough work

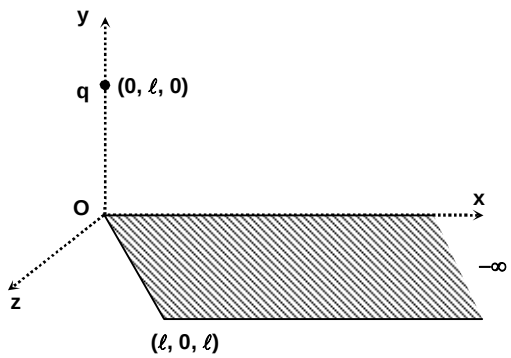
3. A charge particle 'q' lies at the centre of two concentric hollow sphere of inner radii R and 3R and outer radii 2R and 4R respectively. What amount of work has to be performed to slowly transfer the charge 'q' from center through the orifice to infinity.



(A) $\frac{5}{96}\frac{q^2}{\pi\varepsilon_0 R}$ (B) $\frac{q^2}{16\pi\varepsilon_0 R}$

(C) $\frac{7}{96}\frac{q^2}{\pi\varepsilon_0 R}$ (D) $\frac{q^2}{96\pi\varepsilon_0 R}$

4. A point charge 'q' is placed at a distance 'ℓ' (on the y-axis) vertically above the surface which lies in the xz plane as shown in the figure. The flux of electric field passing through the plate is



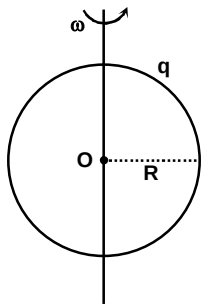
(A) $\frac{q}{6\varepsilon_0}$

(B) $\frac{q}{18\varepsilon_0}$

(C) $\frac{q}{24\varepsilon_0}$

(D) $\frac{q}{48\varepsilon_0}$

5. A hollow non conducting sphere of radius R has a positive charge q uniformly distributed on its surface. The sphere starts rotating with a constant angular velocity 'ω' about an axis passing through center of sphere, as shown in the figure. Then the net magnetic field at center of the sphere is



(A) $\frac{\mu_0 q\omega}{2\pi R}$ (B) $\frac{\mu_0 q\omega}{3\pi R}$

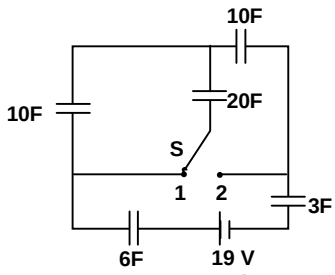
(C) $\frac{\mu_0 q\omega}{6\pi R}$ (D) $\frac{\mu_0 q\omega}{12\pi R}$

Space for Rough work

(One or More than one correct type)

This section contains **EIGHT** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

6. When the steady state is reached by keeping switch in position - 1, then switch 'S' is transferred from position 1 to 2. Then



(A) Charge on 3F capacitor is 30 C after S is transferred from position 1 to 2.

(B) Charge on 3F capacitor is $\frac{650}{19}$ C after S is transferred from position 1 to 2.

(C) Work done by the battery is 380 Joule after S is transferred from position 1 and 2.

(D) Work done by the battery is 80 Joule after S is transferred from position 1 and 2.

7. A cubical box of side 5 m contains helium gas at a pressure 320 N/m^2. During an observation time of 1 second, an atom travelling with root mean square speed parallel to one of the sides of the cube was found to make 1000 hits with a particular wall, without any collision with the other atoms. (take $R = \frac{25}{3}$ J/mol-K). Then

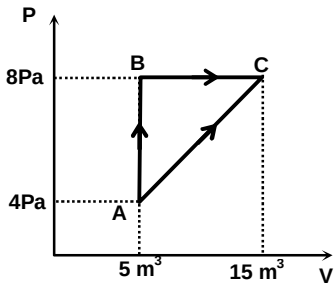
(A) The temperature of gas is 16×10^3 K.

(B) The temperature of gas is 8×10^3 K.

(C) The total mass of helium gas in the box is 1.2 gm.

(D) The total mass of helium gas in the box is 2.4 gm.

8. In the given graph, an ideal gas can change its state from A to C by two paths ABC or AC. Then



(A) If the internal energy of gas at 'A' is 10 J and the amount of heat supplied in path AC is 200 J, then the internal energy of gas at 'C' is 150 J.

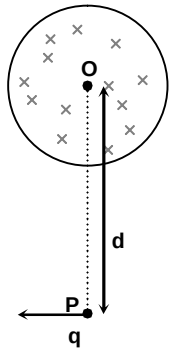
(B) If the internal energy of gas at 'A' is 10 J and the amount of heat supplied in path AC is 200 J, then the internal energy of gas at 'C' is 75 J.

(C) If the internal energy of gas at state B is 20 J. Then the amount of heat supplied to the gas to go from A to B may be 5 Joule.

(D) If the internal energy of gas at state B is 20 J. Then the amount of heat supplied to the gas to go from A to B must be 10 Joule.

Space for Rough work

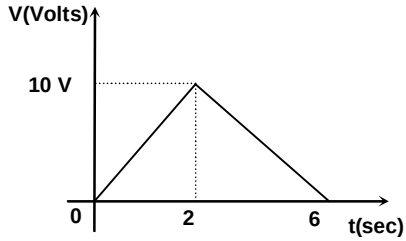
9. In the cylindrical region of radius R = 2m, there exists a time varying magnetic field B such that $\frac{dB}{dt} = 2$ Tesla/sec. A charged particle having charge q = 2C is placed at the point P at a distance d from its center O. Now, the particle is moved in the direction perpendicular to line OP by an external agent up to infinity so that there is no gain in kinetic energy of charged particle. Then



(A) work done by external agent is 4π Joule when d = 4m
(B) work done by external agent is 4π Joule when d = 8m
(C) work done by external agent is independent of d.
(D) work done by external agent is positive.

10. A uniform conducting ring of mass m = 2 kg, radius r = 2m and resistance 8 Ω is kept on smooth, horizontal surface. A time varying magnetic field $\vec{B} = (\hat{i} + t^2\hat{j})$ Tesla is present in the region, where t is time in second and take vertical as y-axis. (Take $\pi^2 = 10$). Then
(A) Time when ring starts toppling is 1 sec.
(B) Time when ring starts toppling is 3/4 sec.
(C) Heat generated through the ring till the instant when the ring start toppling is 40/3 Joule.
(D) Heat generated through the ring till the instant when the ring start toppling is 80/3 Joule.

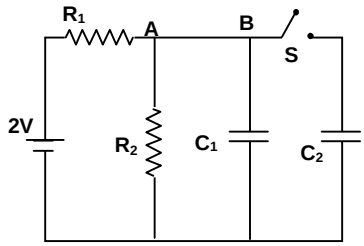
11. The potential difference across a 10 H inductor as a function of time is shown in the figure. At t = 0, current is zero. Then



(A) The current in the inductor at t = 6 sec is 3 amp.
(B) The current in the inductor at t = 1 sec is 1/4 amp.
(C) The current in the inductor at t = 4 sec is 5/2 amp.
(D) The variation of current versus time is parabola.

Space for Rough work

12. Initially the switch 'S' is open for a ling time. Now the switch 'S' is closed at t = 0 ($R_1 = 2\Omega$, $R_2 = 2\Omega$, $C_1 = 1F$ and $C_2 = 1F$). Then



(A) The current through the wire AB at t = 2 sec is 1/2e ampere.

(B) The current through the wire AB at t = 2 sec is 1/e ampere.

(C) The charge on the capacitor C_2, which was initially uncharged, at t = 2 sec is $\left(1-\frac{1}{2e}\right)$ coulombs.

(D) The charge on the capacitor C_1 at t = 2 sec is $\left(1-\frac{1}{2e}\right)$ coulombs.

13. An electric kettle has two coils of different power. When one coil is switched on, it takes 30 minutes to boil a certain amount of water and when second coil is switched on, it takes 45 minutes to boil same amount of water. Then

(A) It takes 150 minutes to boil same amount of water when both the coils are used in series combination.

(B) It takes 75 minutes to boil same amount of water when both the coils are used in series combination.

(C) It takes 18 minutes to boil same amount of water when both the coils are used in parallel combination.

(D) It takes 25 minutes to boil same amount of water when both the coils are used in parallel combination.

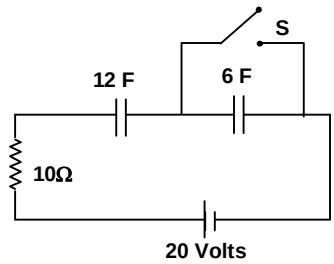
Space for Rough work

SECTION – C
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

14. A ideal gas whose adiabatic exponent equals γ is expanded so that the amount of heat transferred to the gas is equal to twice of decrease of its internal energy. The equation of the process is $TV^{\frac{\gamma-1}{K}} = \text{constant}$ (where T and V are absolute temperature and volume respectively). Then find the value of 'K'.

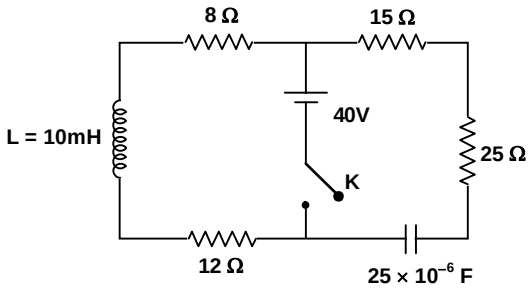
15. When steady state is reached, switch S is opened. Then the amount of heat rejected by the resistor is 800 K Joule. Then find the value of K.



16. A coil of inductance L = 5H and resistance R = 55 Ω is connected in series to the mains alternating voltage of frequency f = 50 Hz in series. What can be the non-zero capacitance of the capacitor (in μF) connected in series with the coil if the power dissipated has to remain unchanged. (take $\pi^2 = 10$)

17. A particle having charge 'q' and mass 'm' is projected with velocity $(4\hat{i} - 6\hat{j} + 3\hat{k})$ m/sec from the origin in a region occupied by electric field 'E' and magnetic field 'B' such that $\vec{B} = B_0\hat{j}$ and $\vec{E} = E_0\hat{j}$ (take $\frac{qE_0}{m} = 2$). Find the time (in sec) when the magnitude of velocity of the charge particle becomes $5\sqrt{5}$ m/sec. (neglect the gravity)

18. In the circuit the key (K) is closed at t = 0. Find the current in ampere through the key(K) at the instant t = $10^{-3}\ell n2$ sec.



Space for Rough work

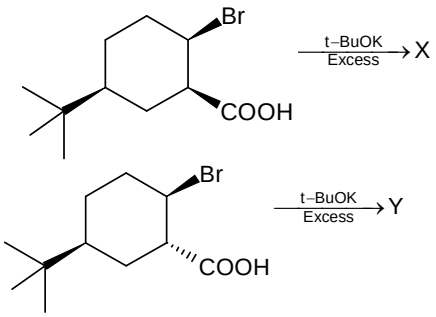
Chemistry

PART – II

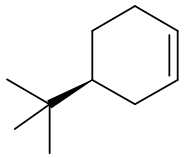
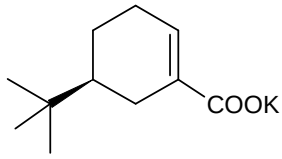
SECTION – A
(One Options Correct Type)

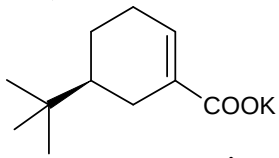
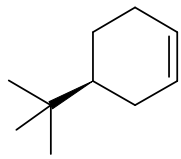
This section contains **5 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

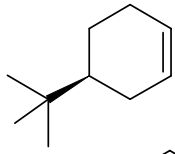
19.

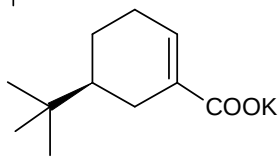


Which of the following option is correct?

(A) X is  : Y is  (Major products)

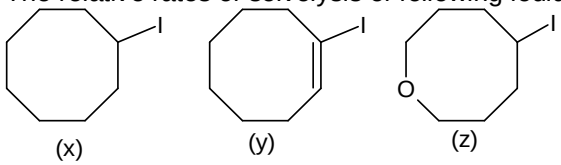
(B) X is  : Y is  (Major products)

(C) Both X & Y are  (Only)

(D) Both X & Y are  (Only)

Space for Rough work

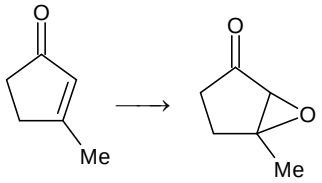
20. The relative rates of solvolysis of following iodides are



(x) (y) (z)

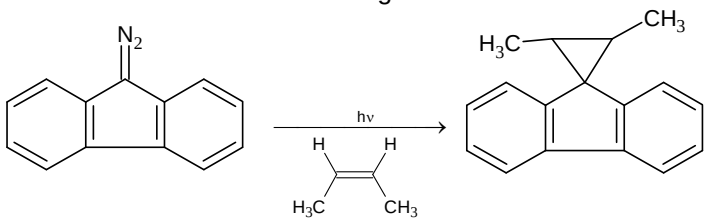
(A) x > y > z
(B) y > x > z
(C) x > z > y
(D) z > x > y

21. The suitable reagent for the following conversion is



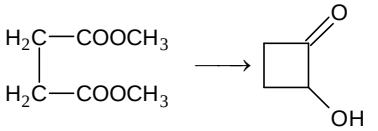
(A) m-CPBA
(B) H_2O_2/AcOH
(C) BuOH/HCl
(D) H_2O_2/NaOH

22. The intermediate in the reaction given below is:



(A) Free radical
(B) Carbocation
(C) Carbanion
(D) Carbene

23. The correct combination of reagents required to effect the following conversion is



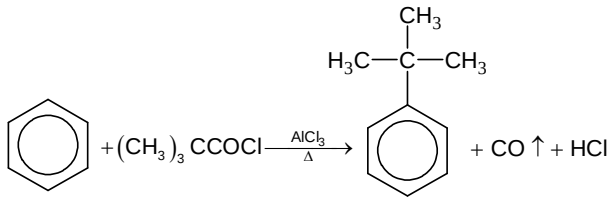
(A) (i) Na, xylene, heat (ii) H_2O_2, NaOH
(B) (i) NaOEt, EtOH (ii) Na, xylene, heat
(C) (i) Na, xylene, Me_3SiCl, heat (ii) H_3O^+
(D) (i) $TiCl_3$, Zn – Cu, $(CH_3)_3SiCl$, heat (ii) H_3O^+

Space for Rough work

(One or More than one correct type)

This section contains **EIGHT** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

24.

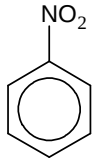


Which of the following statement(s) is/are correct?

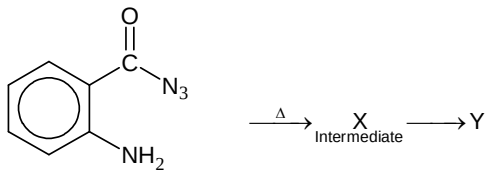
(A) ΔS for the rearrangement of $(CH_3)_3C\overset{+}{C}O$ to $(CH_3)_3\overset{+}{C}$ is positive.

(B) $(CH_3)_3\overset{+}{C}$ is more stable than $(CH_3)_3C\overset{+}{C}O$.

(C) It's an example of homogeneous catalysis.

(D) Solvent used for Friedel Craft acylation is 

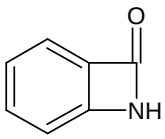
25.

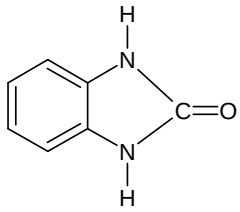


X & Y are:

(A) X is acyl cation

(B) X is nitrene

(C) Y is 

(D) Y is 

Space for Rough work

26. In the reaction,

$$2x + B_2H_6 \longrightarrow \left[BH_2(x)_2\right]^+ \left[BH_4\right]^-$$

The reactant(s) 'x' is (are):

(A) NH_3 (B) CH_3NH_2

(C) $(CH_3)_2NH$ (D) $(CH)_3N$

27. Which of the following are true for Bakelite?

(A) Cross linked polymer (B) Condensation polymer

(C) Thermosetting polymer (D) Addition polymer

28. Which of the following statements about the peptide bond is(are) correct?

(A) The C – N bond of the peptide linkage is longer than that of an amide.

(B) The $-\overset{\overset{O}{\|}}{C}-NH-$ portion of the peptide linkage is planar.

(C) The C – N bond length of peptide is shorter than that of an amine.

(D) The rotation about CO – N bond of peptide is restricted.

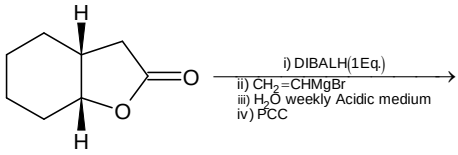
29. In which of the following pairs at least one of the compounds give positive Tollens test?

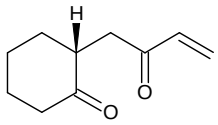
(A) Glucose and sucrose (B) Glucose and fructose

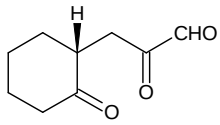
(C) Fructose and sucrose (D) Maltose and sucrose

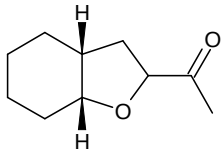
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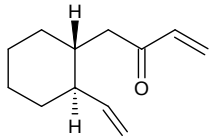
30. Which of the following is/are not the products in the following reaction?



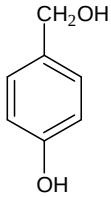
(A) 

(B) 

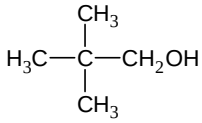
(C) 

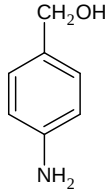
(D) 

31. Which of the following can give aldehyde by the oxidation with Jones reagent?

(A) 

(B) $CH_2 = CH - CH_2OH$

(C) 

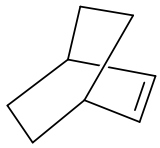
(D) 

Space for Rough work

SECTION – C
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

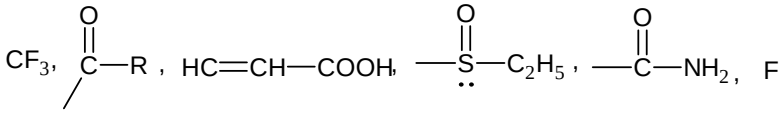
32. Pure (+) 2-chlorobutane gives how many different dichlorbutanes?

33. 

No. of aldol condensation product.

34. Number of amine isomer of the molecular formula $C_4H_{11}N$ is/are structural isomers

35. Number of following substituents those are deactivating but ortho and para directing.



36. Total number of chiral carbons in $\beta-D(+)$ glucose is:

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **5 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. Consider two parabola $y = x^2 - x + 1$ and $y = -x^2 + x + \frac{1}{2}$, the parabola $y = -x^2 + x + \frac{1}{2}$ is fixed and parabola $y = x^2 - x + 1$ rolls without slipping around the fixed parabola, then the locus of the focus of the moving parabola is

(A) $y = \frac{3}{4}$ (B) $y = -\frac{3}{4}$

(C) $y = x^2$ (D) $y = -x^2$

38. Three circles C_1, C_2, C_3 with radii r_1, r_2, r_3 ($r_1 < r_2 < r_3$) respectively are given as $r_1 = 2$, and $r_3 = 8$ they are placed such that C_2 lies to the right of C_1 and touches it externally, C_3 lies to the right of C_2 and touches it externally. There exist two straight lines each of which is a direct common tangent simultaneously to all the three circles then r_2 is equal to

(A) $r_2 = 4$ (B) $r_2 = 5$

(C) $r_2 = 10$ (D) $r_2 = 16$

39. Let A, B, C and D be four distinct point on a line in that order. The circles with diameters AC is $x^2 + y^2 + ax + c = 0$ and BD is $x^2 + y^2 - by = 0$ intersect at X and Y the line XY meets BC at Z. Let P be a point on XY other than Z, the line CP intersects the circle with diameter AC at C and M, and line BP intersects the circle with diameter BD at B and N and the equation of line AM and DN are $bx + cy + a = 0$ and $cx + ay + b = 0$ respectively, then which of the following is true (where ω is a cube root of unity)

(A) $a + b + c = 1$ (B) $a\omega + b\omega^2 + c = 0$

(C) $a + b\omega^2 + c\omega = 0$ (D) none of these

Space for rough work

40. Let A and B be the ends of the diameter $4x - y - 15 = 0$ of the circle $x^2 + y^2 - 6x + 6y - 16 = 0$. A and B lie on tangents at the end points of the major axis of an ellipse such that line joining A and B is a tangent to the same ellipse at a point P. If the equation of major axis of the ellipse is $y = x$ then the distance between the foci is

(A) $2\sqrt{2}$ (B) $4\sqrt{2}$

(C) 8 (D) $2\sqrt{3}$

41. Two circle with radii r_1 and r_2 respectively touch each other externally. Let r_3 be the radius of a circle that touches these two circles as well as a common tangents to two circles then which of the following relation is true

(A) $\frac{1}{\sqrt{r_3}} = \frac{1}{\sqrt{r_1}} + \frac{1}{\sqrt{r_2}}$ (B) $\frac{1}{\sqrt{r_3}} = \left|\frac{1}{\sqrt{r_1}} - \frac{1}{\sqrt{r_2}}\right|$

(C) $\sqrt{r_3} = \sqrt{r_1} + \sqrt{r_2}$ (D) $\sqrt{r_3} = \left|\sqrt{r_1} - \sqrt{r_2}\right|$

(One or More than one correct type)

This section contains **EIGHT** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

42. A point M divides A and B in the ratio 1 : 2 where A and B diametrically opposite ends of a circle $x^2 + y^2 - 5x - 9y + 22 = 0$ square AMCD and BMEF on the length AM and MB are constructed on the same side of line AB if co-ordinates of A is (1, 3) then find the orthocentre of ΔABE

(A) (1, 6) (B) (1, 5)

(C) (3, 3) (D) (4, 6)

43. If largest and smallest value of $\frac{y-4}{x-3}$ is p and q where (x, y) satisfy $x^2 + y^2 - 2x - 6y + 9 = 0$ then which of the following is true

(A) $p + q = \frac{4}{3}$ (B) $q = 1$

(C) $p = \frac{4}{3}$ (D) $pq = \frac{4}{3}$

Space for rough work

44. A particle P moves on the line y = x, $x \geq 0$ with constant speed u m/sec. Another particle Q revolves along the circle $(x-1)^2 + y^2 = \frac{1}{9}$ with constant speed $\frac{\omega}{3}$ m/sec. in anticlockwise direction, let at t = 0, both P and Q simultaneously started, P from the point (0, 0) and Q from the point $\left(\frac{2}{3}, 0\right)$. Further at any time instant t sec., let the distance of their separation is r(t) m. Also it was observed that at time t_0 sec., both P and Q are at their closest distance of approach, denoted by $r_{minimum}$ m, then choose the correct statements from among the following

(A) $[12\, r_{minimum}] = 4$ (where [.] denotes the greatest integer function)

(B) $t_0 = \frac{7\pi}{4\omega}$ sec

(C) $t_0 = \frac{\pi}{2\omega}$ sec

(D) $2r(t)\frac{dr(t)}{dt}\Big|_{t=\frac{7\pi}{2\omega}\text{sec}} = \frac{16\omega}{3\pi}$ m²/sec

45. A rectangle HOMF has sides HO = 11 and OM = 5. A triangle ABC has H as the intersection of the altitude, O the centre of the circumscribed circle, M the mid point of BC, and F the foot of the altitude from A, then

(A) perimeter of ΔABC is greater than 70
(B) area of ΔABC in integer
(C) one side of ΔABC in rational
(D) all sides of ΔABC are less than 30

46. All x in the interval $\left(0, \frac{\pi}{2}\right)$ such that $\frac{\sqrt{3}-1}{\sin x} + \frac{\sqrt{3}+1}{\cos x} = 4\sqrt{2}$ is

(A) $\frac{\pi}{12}$
(B) $\frac{11\pi}{36}$
(C) $\frac{13\pi}{36}$
(D) $\frac{\pi}{6}$

47. In a triangle ABC, $\frac{a}{b} = 2 + \sqrt{3}$, and $\angle C = 60^\circ$ then angle A and B is

(A) $\angle A = 105^\circ$
(B) $\angle B = 15^\circ$
(C) $\angle A = 115^\circ$
(D) $\angle B = 105^\circ$

Space for rough work

48. Let cos A + cos B = x; cos 2A + cos 2B = y; cos 3A + cos 3B = z, then which of the following is true

(A) $\cos^2 A + \cos^2 B = 1 + \frac{y}{2}$ (B) $\frac{1}{4}\left(2x^2 - y - 2\right) = \cos A \cos B$

(C) $2x^2 + z = 3x(1 + y)$ (D) $xyz = 0 \ \forall \ A, B \in R$

49. Let A(4, 3), B(–4, 3) and C(0, –5) be the vertices of triangle and P(5, 0) let L, M and N be the feet of the perpendiculars drawn from P onto the sides BC, CA and AB respectively, then

(A) area of ΔLMN is 5 sq. units (B) the centroid of ΔLMN does not exist

(C) the orthocentre of ΔLMN is the origin (D) L, M and N are collinear

SECTION – C
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

50. The number of solution in the interval $[0, \pi]$ of the equation $\sin^3 x \cdot \cos 3x + \sin 3x \cdot \cos^3 x = 0$ is _____

51. The number of values of θ in the range $[0, 2\pi]$ satisfying the equation $\cos^4 2\theta + 2\sin^2 2\theta = 17(\sin\theta + \cos\theta)^8$ is _____

52. A line divides an equilateral triangle ABC of side 1 unit into two parts with same perimeter and different area S_1 and S_2 sq. units. If maximum value of $\frac{S_1}{S_2} = \frac{m}{n}$ (where m and n are coprime), then $|m - n|$ is _____

53. If the tangent at the point (h, 2) on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ cuts the circle $x^2 + y^2 = a^2$ at the points $Q(x_1y_1)$ and $R(x_2y_2)$, then the value of $\frac{1}{y_1} + \frac{1}{y_2}$ is _____

54. In a triangle ABC, a = 3, b = 4, c = 5, then the value of $\frac{rr_1r_2r_3}{6}$ is _____

Space for rough work